

Education

Rutgers University, New Brunswick

- PhD** *Machine Learning* - 4.0 GPA Sept 2024 – Present
- **Coursework:** Reinforcement Learning, Machine Vision, Robotics Design
- BS** *Electrical & Computer Engineering* - 3.79 GPA, *Summa Cum Laude* Sept 2019 – May 2024

Experience

- Rutgers University**, Graduate Research Assistant New Brunswick, NJ
March 2024 – Present
- NSF NRT Funded Fellow in the SOCRATES (Socially Cognizant Robotics for a Technologically Enhanced Society) program from 2024-2026
 - Currently researching egocentric video modeling and scene representations for robot navigation
 - Developed systems for autonomous data collection for agricultural applications
- SRI International**, Engineering Intern Princeton, NJ
Summers 2022, 2023
- Conducted testing of CCD and LIDAR imaging devices
 - Analyzed test data for imaging devices, including wafer yield and die performance

Publications and Projects

- R. John**, A. Kesari, V. DiMatteo, K. Dana. "EgoCampus: Egocentric Pedestrian Eye Gaze Model and Dataset", *Arxiv preprint*
- Collected a egocentric pedestrian video dataset with eye gaze information; over 30 hours of data.
 - Designed and trained a modular network for gaze-prediction; combines image features and video features for final prediction.
- R. John**, et al. "Modeling Time-Lapse Trajectories to Characterize Cranberry Growth", *ICCV workshop on computer vision in plant phenotyping and agriculture (CVPPA 2025)*
- Learned interpretable latent spaces by modeling plant growth
 - Accurate predictions on plant variety and growth period
 - Contributes a time-lapse image dataset with multiple varieties of cranberries

Extracurriculars and Outreach

- Rutgers Novice to Expert Robotics Club** Mentor and organizer Sep 2025 - May 2026
- Organized and led sessions mentoring undergraduates on topics in robotics
- Rutgers University ECE Capstone Design** Awarded 8th place and tie in best social impact May 2024
- VR technology and motion sensors for personalized physical therapy
- Rutgers Intelligent Ground Vehicle Competition** Division lead Sep 2023 - May 2024
- Led team developing electronics suite for robot in autonomous navigation challenge

Presentations

- April 2026 **Poster Presenter** Princeton Robotics Symposium
- Oct '25,'26 **Student Paper Presenter** Rutgers Robotics Workshop
- Oct 2025 **Lightning Talk Presenter** ICCV Workshop on Plant Phenotyping and Agriculture (CVPPA)
- Feb '25, Apr '26 **Poster Presenter** NYC Computer Vision Day

Skills

Languages: Python, Rust, C, Java, MATLAB

Technologies: Git, Linux, Anaconda, Pytorch, Numpy, Pandas, HuggingFace, Matplotlib, WebGPU, ROS, Rerun